

Gate (II.11), phase V1:

Trace representations with a marked slice, the free thermal Weyl functionals,
and the thermal fixed-coarse rate

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Abstract

This document executes phase V1 of the thermal two-torus route (`gate_III11_scoping.tex`): Statements A2(a) and A3 are proved in full, and A2(b) is put into its final conditional form. Contents: the Schrödinger realization of the lattice Weyl operators with the exact CCR phase; trace-class bounds and eigenfunction estimates giving the kernel identity for $\text{Tr}(e^{-tK_{N,L}} W_N(\xi))$ with complete Fubini justification; the exact *mean-shift representation* of the marked-slice trace — the momentum component of ξ enters through an explicit harmonic jump interpolant H_p which satisfies the discrete Mehler-chain harmonicity *exactly* at every time discretization, so the Gaussian change of variables is exact, its constant is the classical action of the interpolant — precisely the momentum half of the thermal covariance form — and the shift phase cancels the CCR phase identically; the closed free thermal Weyl functionals on both sides (lattice: finite mode products; continuum: countable products with a complete convergence argument), by an elementary Laguerre-series proof of the single-mode thermal displacement formula; and the thermal fixed-coarse rate theorem, uniform on $t \in [t_0, \infty]$, obtained from the proved vacuum-rate lemmas with two new derivative bounds and explicit constants. Every import is isolated with its hypotheses; the V0 numeric tables verify each closed form proved here.

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1 Conventions, imports, and the proof contract

We work in the setting of the Part II model sheet (`lamport_part_ii_model_sheet.tex`), whose conventions are used without change: the spatial torus $\mathbb{T}_L = \mathbb{R}/(2L\mathbb{Z})$; the lattice Λ_N of $2r_N$ sites with spacing $\varepsilon_N = L/r_N$ and momenta $\Gamma_N = \{\pi j/L : -r_N \leq j \leq r_N - 1\}$; the dispersion $\gamma_N(k) = (m^2 + 4\varepsilon_N^{-2} \sin^2(\varepsilon_N k/2))^{1/2}$; the canonical pairs $[\Phi_N(x), \Pi_N(y)] = i\varepsilon_N^{-1} \delta_{xy}$; the smeared fields $\Phi_N(q) = \varepsilon_N \sum_x q(x) \Phi_N(x)$, $\Pi_N(p) = \varepsilon_N \sum_x p(x) \Pi_N(x)$; the Weyl operators $W_N(\varepsilon_N q + ip) = e^{i(\Phi_N(q) + \Pi_N(p))}$; the hat $\hat{f}(k) = \varepsilon_N^{1/2} \sum_x f(x) e^{-ikx}$; the free Hamiltonian $H_{N,L}^{(0)}$ of model-sheet (3.1); the Wick covariance $c_{N,L}$ of (3.4); and the interacting operator

$$K_{N,L} = H_{N,L}^{(0)} + V_N, \quad V_N := \lambda \varepsilon_N \sum_{x \in \Lambda_N} : \Phi_N(x)^4 :_{c_{N,L}}, \quad \lambda \geq 0,$$

with its coercive closed-form realization, compact resolvent, and simple ground state (model-sheet Proposition, §3). Everything below except §6 is finite-dimensional in the number of sites (though infinite-dimensional as quantum mechanics), and no constructive-QFT input is used.

Imports. (I1) The *harmonic-reference* Feynman–Kac formula for $e^{-tK_{N,L}}$: jointly continuous, strictly positive kernel

$$e^{-tK_{N,L}}(\mathbf{x}, \mathbf{y}) = k_t^{(0)}(\mathbf{x}, \mathbf{y}) \mathbb{E}_{\mathbf{x}, \mathbf{y}}^t \left[\exp \left(- \int_0^t V_N(B_s) ds \right) \right], \quad (1)$$

where $k_t^{(0)}$ is the kernel of $e^{-tH_{N,L}^{(0)}}$ (the Mehler product of Lemma 5.1), $\mathbb{E}_{\mathbf{x}, \mathbf{y}}^t$ the corresponding *oscillator-bridge* expectation, and V_N the *interaction* alone — not the full multiplication potential: the harmonic part is carried by the reference kernel. Proof: V3 (`v3_continuum_package.tex`), Theorem 3.1(ii), applied with the lattice dictionary of its Remark 3.2 (dispersions $\gamma_N(k)$, $k \in \Gamma_N$; interaction V_N ; class membership by the model-sheet §3 proposition) — the Trotter/monotone-convergence argument there is run against the Mehler reference. *Two bookkeeping remarks.* (a) V3, Theorem 3.1(ii) states the identity in the *Gaussian* (μ -) representation, with the normalized Mehler operator M_t ($M_t \mathbf{1} = \mathbf{1}$); (1) is its Lebesgue form, obtained by the ground-state (Doob) conjugation

$$k_t^{(0)}(\mathbf{x}, \mathbf{y}) = e^{-tE_N^{(0)}} \Omega_0(\mathbf{x}) \Omega_0(\mathbf{y}) M_t(\mathbf{x}, \mathbf{y}), \quad E_N^{(0)} = \frac{1}{2} \sum_{k \in \Gamma_N} \gamma_N(k),$$

Ω_0 the free ground state; the two bridge measures induced on paths coincide, because the interior factors Ω_0^2 are precisely the reference density, and the boundary factors cancel in every ratio used below. (Lemma 5.1 is stated in the rescaled coordinates $\mathbf{X} = \varepsilon_N^{1/2} \boldsymbol{\varphi}$; as a $d\boldsymbol{\varphi}$ -kernel it carries the constant Jacobian $\varepsilon_N^{r_N}$, which likewise cancels in ratios.) (b) *Acyclicity.* V3's Theorem 3.1(ii) is proved from the Mehler kernel, the elementary Trotter product formula for a bounded perturbation, and monotone convergence; its only contact with this document is Lemma 5.1, whose own proof (heat equation plus an L^2 uniqueness argument) uses nothing from (I1), and which V3 re-verifies inline from the Hermite eigenvalues. The

dependency order is therefore Lemma 5.1 $\rightarrow$ V3 3.1(ii) $\rightarrow$ §3 (Lemma 3.2, Theorem 3.3), Lemmas 5.2–5.3, Theorem 5.4 $\rightarrow$ V3 3.1(v)–(vi); no cycle arises. The model sheet’s own formula (3.16) — flat Gaussian kernel, Brownian bridge, *full* potential — is a different identity and is *not* the one used below. (I2) From the free-rate document (`lamport_part_ii_free_fixed_coarse_rate.tex`): the uniform $D4$ envelope (its Lemma 2.1), the filter-tail bound (Lemma 2.2), the dispersion estimates (Lemma 3.1), its pulled-back covariances (Lemmas 1.1 and 1.2), and — in §7 — the *proof architecture* of its Theorem 4.1: the constants $r_*, A_q, A_p, b, \kappa, c_0$ of (4.1)–(4.6) together with C_D of (2.3) and J_r of (4.10), the split (4.10)–(4.11), the counting bounds (4.16) and the integral-comparison tail (4.20)–(4.21), which §7 re-runs verbatim with thermal weights. (I0) The model sheet, including its §3 Proposition (class membership of the finite-torus systems), and the scoping note’s Lemma 2.1 where cited. (I3) Elementary classical facts proved inline where used (Mehler kernel: verified, not imported; Laguerre summation: proved). No other input enters.

Contract. Every limit names its topology; every interchange of sum and integral is dominated explicitly; statements consumed from V0 are numerical verifications only and never premises.

2 The Schrödinger realization and the CCR phase

On $\mathcal{H}_N = L^2(\mathbb{R}^{\Lambda_N}, d\varphi)$ the model sheet realizes $\Phi_N(x)$ as multiplication by φ_x and $\Pi_N(x) = -i\varepsilon_N^{-1}\partial_{\varphi_x}$. Then $\Pi_N(p) = -i\sum_x p(x)\partial_{\varphi_x}$.

Proposition 2.1 (translations and the exact phase). (i) $\Phi_N(q) + \Pi_N(p)$ is essentially self-adjoint on $C_c^\infty(\mathbb{R}^{\Lambda_N})$, and

$$(W_N(\varepsilon_N q + ip)\psi)(\varphi) = e^{i\varepsilon_N q \cdot \varphi + \frac{i}{2}\varepsilon_N q \cdot p} \psi(\varphi + p), \quad (2)$$

where $q \cdot \varphi := \sum_x q(x)\varphi_x$ etc. (ii) Consequently

$$W_N(\varepsilon_N q + ip) = e^{\frac{i}{2}\varepsilon_N q \cdot p} e^{i\Phi_N(q)} e^{i\Pi_N(p)}, \quad (e^{i\Pi_N(p)}\psi)(\varphi) = \psi(\varphi + p). \quad (3)$$

Proof. Define, for $s \in \mathbb{R}$,

$$(V(s)\psi)(\varphi) := e^{is\varepsilon_N q \cdot \varphi + \frac{i}{2}s^2\varepsilon_N q \cdot p} \psi(\varphi + sp).$$

Each $V(s)$ is unitary, $V(s + s') = V(s)V(s')$ by direct substitution (the cross terms ss' combine as $s'\varepsilon_N q \cdot (\varphi) + s\varepsilon_N q \cdot (\varphi + s'p) + \frac{\varepsilon_N}{2}(s^2 + s'^2)q \cdot p = (s + s')\varepsilon_N q \cdot \varphi + \frac{\varepsilon_N}{2}(s + s')^2 q \cdot p$), and $s \mapsto V(s)\psi$ is continuous for $\psi \in C_c^\infty$, hence, by the group property and density, strongly continuous everywhere. For $\psi \in C_c^\infty$,

$$\left. \frac{d}{ds} \right|_{s=0} V(s)\psi = (i\varepsilon_N q \cdot \varphi + p \cdot \nabla)\psi = i(\Phi_N(q) + \Pi_N(p))\psi.$$

By Stone’s theorem V has a self-adjoint generator G extending $(\Phi_N(q) + \Pi_N(p))|_{C_c^\infty}$; since $V(s)$ maps C_c^∞ into itself, C_c^∞ is a core for G (Nelson’s invariant-domain criterion), i.e. $\Phi_N(q) + \Pi_N(p)$ is essentially self-adjoint there and $e^{i(\Phi_N(q) + \Pi_N(p))} = V(1)$, which is (2). Setting $q = 0$ gives the translation formula; comparing (2) with the composition $e^{i\Phi_N(q)}e^{i\Pi_N(p)}$ (multiplication, then translation) gives (3). $\square$

Remark 2.2. The phase $\theta(q, p) = \frac{1}{2}\varepsilon_N q \cdot p$ and the Weyl relation with the model-sheet symplectic form were verified numerically on truncation-safe corners in V0, with residuals 3.9×10^{-15} (CCR phase) and 2.7×10^{-15} (Weyl relation).

3 Trace class, eigenfunction bounds, and the kernel identity

Throughout this section $t > 0$ and $K := K_{N,L}$ with eigenvalues $E_0 < E_1 \leq E_2 \leq \dots \rightarrow \infty$ (compact resolvent; simplicity of E_0 per the model sheet) and real orthonormal eigenfunctions ψ_j .

Proposition 3.1 (trace class). (i) *The pointwise bound $:u^4:_c = u^4 - 6cu^2 + 3c^2 = (u^2 - 3c)^2 - 6c^2 \geq -6c^2$ gives $V_N \geq -12\lambda L c_{N,L}^2 =: -c_V$.* (ii) $E_j(K) \geq E_j(H^{(0)}) - c_V$ for all j , and hence

$$Z_N(t) := \text{Tr } e^{-tK} \leq e^{tc_V} \text{Tr } e^{-tH^{(0)}} = e^{tc_V} \prod_{k \in \Gamma_N} (2 \sinh(t\gamma_N(k)/2))^{-1} < \infty.$$

Proof. (i) is the displayed algebra summed over the $2r_N$ sites with weight $\lambda \varepsilon_N$ ($\varepsilon_N \cdot 2r_N = 2L$). (ii): $K \geq H^{(0)} - c_V$ as quadratic forms on the common form core, so the min-max values obey the same inequality; both operators have discrete spectrum, and the free trace is the standard oscillator product (computed independently in Theorem 4.3 below with its zero-point factors). $\square$

Lemma 3.2 (eigenfunction bounds). *For every $s > 0$ and φ ,*

$$|\psi_j(\varphi)| \leq e^{sE_j} (e^{-2sK}(\varphi, \varphi))^{1/2}, \quad \int e^{-2sK}(\varphi, \varphi) d\varphi = \text{Tr } e^{-2sK} < \infty.$$

Proof. From $\psi_j = e^{sE_j} e^{-sK} \psi_j$ and Cauchy–Schwarz,

$$|\psi_j(\varphi)| \leq e^{sE_j} \|e^{-sK}(\varphi, \cdot)\|_2 = e^{sE_j} (e^{-2sK}(\varphi, \varphi))^{1/2},$$

the last step by the semigroup property and symmetry of the (real) kernel. For the integral: $e^{-2sK} = (e^{-sK})^2$ with e^{-sK} Hilbert–Schmidt (Proposition 3.1) and continuous kernel (1), so

$$e^{-2sK}(\varphi, \varphi) = \int |e^{-sK}(\varphi, \mathbf{y})|^2 d\mathbf{y}, \quad \int e^{-2sK}(\varphi, \varphi) d\varphi \stackrel{\text{Tonelli}}{=} \|e^{-sK}\|_{\text{HS}}^2 = \text{Tr } e^{-2sK}.$$

$\square$

Theorem 3.3 (kernel identity for the marked-slice trace). *For all real q, p and $t > 0$, with $\theta = \frac{1}{2}\varepsilon_N q \cdot p$,*

$$\text{Tr}(e^{-tK} W_N(\varepsilon_N q + ip)) = e^{i\theta} \int_{\mathbb{R}^{\Lambda_N}} e^{-tK}(\varphi + p, \varphi) e^{i\varepsilon_N q \cdot \varphi} d\varphi, \quad (4)$$

both sides absolutely convergent.

Proof. By the spectral decomposition and boundedness of $W := W_N(\varepsilon_N q + ip)$, $\text{Tr}(e^{-tK} W) = \sum_j e^{-tE_j} \langle \psi_j, W \psi_j \rangle$, absolutely convergent since $|\langle \psi_j, W \psi_j \rangle| \leq 1$. By (2),

$$\langle \psi_j, W \psi_j \rangle = e^{i\theta} \int \psi_j(\varphi) e^{i\varepsilon_N q \cdot \varphi} \psi_j(\varphi + p) d\varphi.$$

To interchange $\sum_j$ and $\int$, dominate using Lemma 3.2 with $s = t/4$:

$$\sum_j e^{-tE_j} |\psi_j(\varphi)| |\psi_j(\varphi + p)| \leq \left(\sum_j e^{-tE_j/2} \right) D(\varphi)^{1/2} D(\varphi + p)^{1/2}, \quad D := e^{-tK/2}(\cdot, \cdot)|_{\text{diag}},$$

and $\int D(\varphi)^{1/2} D(\varphi + p)^{1/2} d\varphi \leq (\int D)^{1/2} (\int D(\cdot + p))^{1/2} = \text{Tr } e^{-tK/2} < \infty$ by Cauchy–Schwarz and Lemma 3.2. Fubini then gives (4) with $e^{-tK}(\mathbf{x}, \mathbf{y}) = \sum_j e^{-tE_j} \psi_j(\mathbf{x}) \psi_j(\mathbf{y})$, which agrees with the continuous kernel (1) almost everywhere, hence (both sides being continuous in $(\mathbf{x}, \mathbf{y})$ — the series converges locally uniformly by the same domination) everywhere. $\square$

4 The free thermal Weyl functionals (A3(a), lattice)

4.1 Mode diagonalization

Introduce the rescaled pairs $X_x := \varepsilon_N^{1/2} \Phi_N(x)$, $P_x := \varepsilon_N^{1/2} \Pi_N(x)$, so $[X_x, P_y] = i\delta_{xy}$, and the unitary discrete Fourier transform $\tilde{f}(k) := (2r_N)^{-1/2} \sum_x f(x) e^{-ikx}$. With $\tilde{X}(k), \tilde{P}(k)$ defined likewise (so $\tilde{X}(k)^* = \tilde{X}(-k)$), the model-sheet free Hamiltonian becomes

$$H^{(0)} = \frac{1}{2} \sum_{k \in \Gamma_N} (\tilde{P}(k)^* \tilde{P}(k) + \gamma_N(k)^2 \tilde{X}(k)^* \tilde{X}(k)) = \sum_{k \in \Gamma_N} \gamma_N(k) (a_k^* a_k + \frac{1}{2}),$$

where $a_k := (2\gamma_N(k))^{-1/2} (\gamma_N(k) \tilde{X}(k) + i\tilde{P}(k))$ satisfy $[a_k, a_{k'}^*] = \delta_{kk'}$, $[a_k, a_{k'}] = 0$ (direct computation from $[\tilde{X}(k), \tilde{P}(k')^*] = i\delta_{kk'}$), and

$$\tilde{X}(k) = (2\gamma_N(k))^{-1/2} (a_k + a_{-k}^*), \quad \tilde{P}(k) = -i(\gamma_N(k)/2)^{1/2} (a_k - a_{-k}^*). \quad (5)$$

Lemma 4.1 (displacement amplitudes). *For real q, p , $\Phi_N(q) + \Pi_N(p) = \sum_{k \in \Gamma_N} (\bar{z}_k a_k + z_k a_k^*)$ with*

$$z_k = \varepsilon_N^{1/2} \left[(2\gamma_N(k))^{-1/2} \tilde{q}(k) + i(\gamma_N(k)/2)^{1/2} \tilde{p}(k) \right],$$

$$\sum_{k \in \Gamma_N} |z_k|^2 = \frac{1}{2 \cdot 2r_N} \sum_{k \in \Gamma_N} \left(\frac{|\tilde{q}(k)|^2}{\gamma_N(k)} + \gamma_N(k) |\tilde{p}(k)|^2 \right).$$

Proof. $\Phi_N(q) + \Pi_N(p) = \varepsilon_N^{1/2} \sum_x (q_x X_x + p_x P_x) = \varepsilon_N^{1/2} \sum_k (\overline{\tilde{q}(k)} \tilde{X}(k) + \tilde{p}(k) \tilde{P}(k))$ by Parseval (reality of q, p gives $\tilde{q}(-k) = \overline{\tilde{q}(k)}$). Substituting (5) and collecting the coefficient of a_k (using the $k \rightarrow -k$ symmetry of the sums) gives the displayed z_k . Then

$$|z_k|^2 = \varepsilon_N \left[\frac{|\tilde{q}(k)|^2}{2\gamma_N(k)} + \frac{\gamma_N(k)}{2} |\tilde{p}(k)|^2 + \text{Im}(\tilde{q}(k) \overline{\tilde{p}(k)}) \right],$$

and the imaginary part is odd under $k \rightarrow -k$ ($\tilde{q}(-k) \overline{\tilde{p}(-k)} = \overline{\tilde{q}(k)} \tilde{p}(k)$), so it cancels in the sum; finally $\varepsilon_N |\hat{f}(k)|^2 = |\tilde{f}(k)|^2 / (2r_N)$. $\square$

4.2 The single-mode thermal displacement formula

Lemma 4.2 (Laguerre summation). *For one mode ($[a, a^*] = 1$, number basis $|n\rangle$), $z \in \mathbb{C}$, and $0 < w < 1$: (i) $\langle n, e^{za^*} e^{-\bar{z}a} n \rangle = L_n(|z|^2) := \sum_{j=0}^n \binom{n}{j} \frac{(-|z|^2)^j}{j!}$; (ii) $\sum_{n \geq 0} w^n L_n(x) = (1-w)^{-1} e^{-xw/(1-w)}$, absolutely convergent; (iii) for $H = \gamma(a^*a + \frac{1}{2})$ and the displacement $D(z) = e^{za^* - \bar{z}a}$,*

$$\frac{\text{Tr}(e^{-tH} D(z))}{\text{Tr} e^{-tH}} = \exp \left[-\frac{1}{2} |z|^2 \coth\left(\frac{t\gamma}{2}\right) \right].$$

Proof. (i) $e^{-\bar{z}a} |n\rangle = \sum_{j \leq n} \frac{(-\bar{z})^j}{j!} \sqrt{\frac{n!}{(n-j)!}} |n-j\rangle$ and likewise for the bra with z ; orthonormality gives $\sum_j \frac{(-|z|^2)^j}{(j!)^2} \frac{n!}{(n-j)!} = L_n(|z|^2)$. (ii) Insert (i) and interchange the (absolutely convergent: majorant $\sum_j \frac{x^j}{j!} \sum_n \binom{n}{j} w^n < \infty$) double sum:

$$\sum_n w^n \sum_j \binom{n}{j} \frac{(-x)^j}{j!} = \sum_j \frac{(-x)^j}{j!} \frac{w^j}{(1-w)^{j+1}} = \frac{1}{1-w} e^{-xw/(1-w)},$$

using $\sum_{n \geq j} \binom{n}{j} w^n = w^j (1-w)^{-j-1}$. (iii) $D(z) = e^{-|z|^2/2} e^{za^*} e^{-\bar{z}a}$ (BCH with central $[za^*, -\bar{z}a] = |z|^2$), so with $w = e^{-t\gamma}$ (zero-point factors cancel in the ratio),

$$\frac{\text{Tr}(e^{-tH} D(z))}{\text{Tr} e^{-tH}} = e^{-|z|^2/2} (1-w) \sum_n w^n L_n(|z|^2) = e^{-|z|^2(\frac{1}{2} + \frac{w}{1-w})} = e^{-\frac{|z|^2}{2} \frac{1+w}{1-w}},$$

and $(1+w)/(1-w) = \coth(t\gamma/2)$. $\square$

4.3 The lattice closed form

Theorem 4.3 (A3(a), lattice). *For every $t \in (0, \infty]$ and $\xi = \varepsilon_N q + ip$,*

$$F_N^{(0)}(t; \xi) := \frac{\text{Tr}(e^{-tH_{N,L}^{(0)}} W_N(\xi))}{\text{Tr}(e^{-tH_{N,L}^{(0)}})} = \exp\left[-\frac{1}{4} Q_N^{(t)}(\xi)\right], \quad (6)$$

$$Q_N^{(t)}(\xi) = \frac{1}{2r_N} \sum_{k \in \Gamma_N} \coth\left(\frac{t\gamma_N(k)}{2}\right) \left(\gamma_N(k)^{-1} |\widehat{q}(k)|^2 + \gamma_N(k) |\widehat{p}(k)|^2\right), \quad (7)$$

with $\coth(\infty) := 1$; moreover $\text{Tr} e^{-tH^{(0)}} = \prod_k (2 \sinh(t\gamma_N(k)/2))^{-1}$, and $t = \infty$ reproduces the model-sheet vacuum functional (3.3).

Proof. $\mathcal{H}_N$ factorizes unitarily over the modes, $H^{(0)} = \sum_k \gamma_k (a_k^* a_k + \frac{1}{2})$ with commuting summands, and by Lemma 4.1 and the multimode BCH (all commutators central), $W_N(\xi) = \prod_k D_k(iz_k)$ with $|iz_k| = |z_k|$. The trace factorizes over the finite tensor decomposition; Lemma 4.2(iii) applies per mode:

$$F_N^{(0)}(t; \xi) = \prod_{k \in \Gamma_N} e^{-\frac{1}{2} |z_k|^2 \coth(t\gamma_k/2)},$$

and inserting Lemma 4.1's sum (with the coth weight carried along; the cross terms cancel in the same $k \rightarrow -k$ pairing because $\gamma_N(-k) = \gamma_N(k)$ and hence the weights are even) gives (6)–(7). The partition function is the product of the geometric sums $e^{-t\gamma/2} (1 - e^{-t\gamma})^{-1} = (2 \sinh(t\gamma/2))^{-1}$. As $t \rightarrow \infty$, $\coth \rightarrow 1$ monotonically and the thermal state converges to the vacuum in trace norm, giving the vacuum case. $\square$

Remark 4.4 (V0 acceptance). (6)–(7) are the formulas verified in V0: against exact diagonalization to 6×10^{-10} ($t = 1$) and 2×10^{-16} ($t = 2$), and against the independent position-space matrix-function evaluation to 10^{-9} for $r_N \in \{8, 64\}$, all t .

5 The jump representation and the exact mean shift (A2(a))

This section proves the interacting marked-slice representation. Fix $t > 0$ and write $\omega(k) := \gamma_N(k)$ for the mode frequencies, $n := 2r_N$.

5.1 The free kernel in Mehler form

Lemma 5.1 (Mehler kernel, verified). *In the rescaled coordinates $\mathbf{X} = \varepsilon_N^{1/2} \boldsymbol{\varphi}$ and mode variables, the free kernel factorizes over $k \in \Gamma_N$ into one-dimensional kernels (for $H = \frac{1}{2}(P^2 + \omega^2 X^2)$, which is $\gamma(a^* a + \frac{1}{2})$ per mode)*

$$k_\Delta^{\text{ld}}(x, y) = \left(\frac{\omega}{2\pi \sinh(\omega\Delta)}\right)^{1/2} \exp\left[-\frac{A_\Delta}{2}(x^2 + y^2) + B_\Delta xy\right], \quad (8)$$

$$A_\Delta = \omega \coth(\omega\Delta), \quad B_\Delta = \frac{\omega}{\sinh(\omega\Delta)}.$$

Proof. Call the right side κ_Δ . A direct computation gives $\partial_\Delta \kappa_\Delta = (\frac{1}{2}\partial_x^2 - \frac{\omega^2}{2}x^2)\kappa_\Delta$ (using $A' = -B^2 \dots$ explicitly: $A'_\Delta = -\omega^2/\sinh^2$, $B'_\Delta = -\omega^2 \cosh/\sinh^2$, and the prefactor contributes $-\frac{1}{2}\omega \coth$; collecting terms on both sides matches coefficient by coefficient in $\{1, x^2, y^2, xy\}$), and $\kappa_\Delta \rightarrow \delta$ as $\Delta \downarrow 0$ in the weak sense against C_c (Gaussian concentration with $A_\Delta - B_\Delta = \omega \tanh(\omega\Delta/2) \rightarrow 0$, $A_\Delta + B_\Delta \rightarrow \infty$). Hence for $\psi \in C_c^\infty$, $u(\Delta) := \int \kappa_\Delta(\cdot, y)\psi(y)dy$ is a strongly differentiable L^2 -solution of $u' = -Hu$, $u(0) = \psi$, as is $e^{-\Delta H}\psi$; for the difference v , $\frac{d}{d\Delta}\|v\|^2 = -2\langle v, Hv \rangle \leq 0$ and $v(0) = 0$, so $v \equiv 0$. Thus κ_Δ is the kernel of $e^{-\Delta H}$. $\square$

5.2 The harmonic jump interpolant

Lemma 5.2 (jump interpolant). *Fix $t > 0$, a mode frequency $\omega > 0$, and a jump $\rho \in \mathbb{R}$. There is exactly one $H \in C^\infty([0, t])$ with*

$$-\ddot{H} + \omega^2 H = 0, \quad H(t) - H(0) = \rho, \quad \dot{H}(t) = \dot{H}(0),$$

namely $H(s) = A \cosh(\omega s) + B \sinh(\omega s)$ with

$$A = -\frac{\rho}{2}, \quad B = \frac{\rho}{2} \coth\left(\frac{\omega t}{2}\right). \quad (9)$$

It satisfies, for every $\Delta > 0$ and $s \in [\Delta, t - \Delta]$, the exact three-term recursion

$$H(s + \Delta) + H(s - \Delta) = 2 \cosh(\omega \Delta) H(s), \quad (10)$$

its boundary values are $H(0) = -\rho/2$, $H(t) = +\rho/2$, and its classical action is

$$S(H) := \int_0^t \frac{1}{2}(\dot{H}^2 + \omega^2 H^2) ds = \frac{1}{2} \dot{H}(0) \rho = \frac{\omega \rho^2}{4} \coth\left(\frac{\omega t}{2}\right). \quad (11)$$

Proof. The general solution of $-\ddot{H} + \omega^2 H = 0$ is $A \cosh(\omega s) + B \sinh(\omega s)$. The two conditions read $A(\cosh(\omega t) - 1) + B \sinh(\omega t) = \rho$ and $A \sinh(\omega t) + B(\cosh(\omega t) - 1) = 0$; solving, $A = -B(\cosh - 1)/\sinh$ and $B[\sinh^2 - (\cosh - 1)^2]/\sinh = \rho$, and since $\sinh^2 - (\cosh - 1)^2 = 2(\cosh - 1)$, $B = \rho \sinh(\omega t)/(2(\cosh(\omega t) - 1)) = \frac{\rho}{2} \coth(\omega t/2)$ and $A = -\rho/2$; uniqueness because the homogeneous system ($\rho = 0$) forces $A = B = 0$. (10) is the addition formula $\cosh(\omega(s \pm \Delta))$, $\sinh(\omega(s \pm \Delta))$ summed. For (11): integrate by parts, $S(H) = \frac{1}{2}[H\dot{H}]_0^t = \frac{1}{2}(H(t)\dot{H}(t) - H(0)\dot{H}(0)) = \frac{1}{2}\dot{H}(0)(H(t) - H(0))$ using $\dot{H}(t) = \dot{H}(0)$; and $\dot{H}(0) = \omega B = \frac{\omega \rho}{2} \coth(\omega t/2)$. $\square$

For the lattice, define the vector interpolant $\mathbf{H}_p(s)$ mode by mode: in the rescaled coordinates the translation by p is the shift by $\varepsilon_N^{1/2} p$, whose k -th Fourier component is $\varepsilon_N^{1/2} \tilde{p}(k)$; let $H_k(s)$ be the interpolant of Lemma 5.2 with $\omega = \gamma_N(k)$ and jump $\rho_k = \varepsilon_N^{1/2} \tilde{p}(k)$ (complex ρ : apply the lemma to real and imaginary parts; reality of the resulting position-space path follows from $\rho_{-k} = \bar{\rho}_k$), and let $\mathbf{H}_p$ be the inverse transform. Then, by (11) and $\varepsilon_N |\tilde{p}(k)|^2 = |\hat{p}(k)|^2/(2r_N)$,

$$S_N(\mathbf{H}_p) := \sum_{k \in \Gamma_N} S(H_k) = \frac{1}{4 \cdot 2r_N} \sum_{k \in \Gamma_N} \gamma_N(k) \coth\left(\frac{t\gamma_N(k)}{2}\right) |\hat{p}(k)|^2, \quad (12)$$

which is exactly the momentum half of $\frac{1}{4}Q_N^{(t)}$; and $\mathbf{H}_p(0) = -p/2$, $\mathbf{H}_p(t) = +p/2$ in the original coordinates (the mode relation $H_k(0) = -\rho_k/2$ is linear, so it inverts to $\mathbf{X}$ -shift $-\varepsilon_N^{1/2} p/2$, i.e. φ -shift $-p/2$).

5.3 The exact change of variables

Let $0 = s_0 < s_1 < \dots < s_M = t$ be any partition, $\Delta_i := s_{i+1} - s_i$. Define the (unnormalized) *jump chain integrals*: for bounded measurable $f_i : \mathbb{R}^n \rightarrow \mathbb{C}$,

$$J_p[f_0, \dots, f_{M-1}] := \int \dots \int \prod_{i=0}^{M-1} k_{\Delta_i}^{(0)}(\mathbf{y}_{i+1}, \mathbf{y}_i) \prod_{i=0}^{M-1} f_i(\mathbf{y}_i) d\mathbf{y}_0 \dots d\mathbf{y}_{M-1} \Big|_{\mathbf{y}_M := \mathbf{y}_0 + p},$$

and J^{per} the same with $\mathbf{y}_M := \mathbf{y}_0$ (periodic chain).

Lemma 5.3 (exact shift identity). *For every partition and all bounded measurable f_i ,*

$$J_p[f_0, \dots, f_{M-1}] = e^{-S_N(\mathbf{H}_p)} J^{\text{per}}[f_0(\cdot + \mathbf{H}_p(s_0)), \dots, f_{M-1}(\cdot + \mathbf{H}_p(s_{M-1}))]. \quad (13)$$

Proof. Work per mode in the rescaled Fourier coordinates (the free chain factorizes by Lemma 5.1; the substitution below is a composition of one-dimensional ones). Substitute $\mathbf{y}_i = \boldsymbol{\eta}_i + \mathbf{H}_p(s_i)$ (unit Jacobian). Each Mehler factor transforms by

$$\log k_{\Delta}^{\text{ld}}(\boldsymbol{\eta}' + a', \boldsymbol{\eta} + a) = \log k_{\Delta}^{\text{ld}}(\boldsymbol{\eta}', \boldsymbol{\eta}) + [(-A_{\Delta}a' + B_{\Delta}a)\boldsymbol{\eta}' + (-A_{\Delta}a + B_{\Delta}a')\boldsymbol{\eta}] - S_{\Delta}^{\text{cl}}(a', a),$$

where $S_{\Delta}^{\text{cl}}(a', a) := \frac{A_{\Delta}}{2}(a'^2 + a^2) - B_{\Delta}a'a$ is the classical action of the harmonic path from a to a' in time Δ (read off Lemma 5.1). Sum over the chain with $a_i = H(s_i)$ (and $a_M = H(s_M) = H(s_0) + \rho$, matching $\mathbf{y}_M = \mathbf{y}_0 + p \mapsto \boldsymbol{\eta}_M = \boldsymbol{\eta}_0$, so the chain closes periodically in $\boldsymbol{\eta}$). The linear terms at the node $\boldsymbol{\eta}_i$ ($1 \leq i \leq M-1$) collect to

$$[-A_{\Delta_{i-1}}H(s_i) + B_{\Delta_{i-1}}H(s_{i-1})] + [-A_{\Delta_i}H(s_i) + B_{\Delta_i}H(s_{i+1})] = 0 :$$

indeed, for the *harmonic* H , $H(s_{i\pm 1}) = \cosh(\omega\Delta)H(s_i) \pm \sinh(\omega\Delta)\dot{H}(s_i)/\omega$ (addition formulas), so $-A_{\Delta}H(s_i) + B_{\Delta}H(s_i \mp \Delta) = \mp \dot{H}(s_i)/\dots$ — explicitly, $-\omega \coth(\omega\Delta)H(s_i) + \frac{\omega}{\sinh(\omega\Delta)}[\cosh(\omega\Delta)H(s_i) \mp \frac{\sinh(\omega\Delta)}{\omega}\dot{H}(s_i)] = \mp \dot{H}(s_i)$, and the two contributions ($-$ from the left factor, $+$ from the right factor) cancel. At the glued node $\boldsymbol{\eta}_0 = \boldsymbol{\eta}_M$ the same computation applies with the one-sided derivatives $\dot{H}(0)$ and $\dot{H}(t)$, which are equal by construction, so that node's linear term vanishes as well. The constants collect to $-\sum_i S_{\Delta_i}^{\text{cl}}(H(s_{i+1}), H(s_i))$; since H restricted to $[s_i, s_{i+1}]$ is the classical trajectory between its endpoints, each term equals the true action integral $\int_{s_i}^{s_{i+1}} \frac{1}{2}(\dot{H}^2 + \omega^2 H^2)$, and the sum telescopes to $S(H)$ of (11). Summing over modes gives $e^{-S_N(\mathbf{H}_p)}$ and (13). $\square$

Theorem 5.4 (A2(a): interacting mean-shift representation). *For $t > 0$, real q, p , and $\lambda \geq 0$, with the periodic free ensemble $\mu_{N,t}^{\text{per}}$ (the Gaussian process on the time circle with the covariance of the periodic Mehler chain; Remark 5.5) and $\mathcal{V}_N(\boldsymbol{\varphi}) := \lambda \varepsilon_N \sum_x : \varphi_x^4 :_{c_{N,L}}$ the interaction multiplication function (the harmonic part of the potential is carried by the ensemble; import (I1)), which satisfies the pointwise bound*

$$\mathcal{V}_N \geq -6\lambda \varepsilon_N (2r_N) c_{N,L}^2 = -12\lambda L c_{N,L}^2 = -c_V$$

of Proposition 3.1 (complete the square: $:u^4:_{\text{c}} = (u^2 - 3c)^2 - 6c^2$; this display is deliberately unnumbered, so that the equation numbers cited by V2–V4 are unchanged):

$$\frac{\text{Tr}(e^{-tK_{N,L}} W_N(\varepsilon_N q + ip))}{\text{Tr}(e^{-tK_{N,L}})} = e^{-S_N(\mathbf{H}_p)} \frac{\mathbb{E}_{\mu_{N,t}^{\text{per}}} \left[e^{i\varepsilon_N q \cdot \boldsymbol{\eta}(0)} e^{-\int_0^t \mathcal{V}_N(\boldsymbol{\eta}(s) + \mathbf{H}_p(s)) ds} \right]}{\mathbb{E}_{\mu_{N,t}^{\text{per}}} \left[e^{-\int_0^t \mathcal{V}_N(\boldsymbol{\eta}(s)) ds} \right]}, \quad (14)$$

with $S_N(\mathbf{H}_p)$ as in (12); the CCR phase has cancelled against the shift phase ($e^{i\theta} e^{i\varepsilon_N q \cdot \mathbf{H}_p(0)} = e^{\frac{i}{2}\varepsilon_N q \cdot p} e^{-\frac{i}{2}\varepsilon_N q \cdot p} = 1$). Expanding $\mathcal{V}_N(\cdot + \mathbf{H}_p)$, the interaction changes by the finite Wick binomial

$$\mathcal{U}_N(\boldsymbol{\eta} + \mathbf{H}_p) - \mathcal{U}_N(\boldsymbol{\eta}) = \lambda \varepsilon_N \sum_x \int_0^t \left[4H_x : \eta_x^3 : + 6H_x^2 : \eta_x^2 : + 4H_x^3 \eta_x + H_x^4 \right] ds, \quad (15)$$

(here $\mathcal{U}_N(\boldsymbol{\eta}) := \int_0^t \mathcal{V}_N(\boldsymbol{\eta}(s)) ds$ denotes the time-integrated interaction along a path), a polynomial with smooth bounded deterministic coefficients (by the closed form derived in the proof, for each fixed N , $\|\mathbf{H}_p(s)\|_\infty \leq \frac{1}{2}\|p\|_{\ell^2(\Lambda_N)}$ uniformly in $s \in [0, t]$ and $t \geq t_0$, with no dependence on t_0 or m ; here $\|p\|_{\ell^2(\Lambda_N)}^2 = \sum_x |p(x)|^2$ is the counting-measure norm of the model sheet (1.4), and the constant $\frac{1}{2}$ is attained, since $\mathbf{H}_p(0) = -p/2$). This bound is not uniform in N along the D4 refinement: the mask is orthogonal, $\sum_n h_n h_{n+2j} = \delta_{j0}$, so $\|R_M^{M+r} p_M\|_{\ell^2(\Lambda_{M+r})} = 2^{r/2} \|p_M\|_{\ell^2(\Lambda_M)}$ and the right-hand side diverges. The N -uniform coarse-data bound that the route actually consumes is $\sup_N \|\mathbf{H}_{p_N}\|_\infty \leq \Sigma_M(p)$ of V2, Lemma 6.3(i), which sums the D4 envelope over Γ_N instead of applying Cauchy–Schwarz over $2r_N$ modes; nothing downstream consumes the bound stated here.

Proof. By Theorem 3.3 and the Feynman–Kac formula (1), the numerator trace is

$$e^{i\theta} \int d\boldsymbol{\varphi} e^{i\varepsilon_N q \cdot \boldsymbol{\varphi}} k_t^{(0)}(\boldsymbol{\varphi} + p, \boldsymbol{\varphi}) \mathbb{E}_{\boldsymbol{\varphi}+p, \boldsymbol{\varphi}}^t \left[e^{-\int_0^t \mathcal{V}_N} \right] = e^{i\theta} \lim_{M \rightarrow \infty} J_p \left[e^{i\varepsilon_N q \cdot (\cdot)} e^{-\Delta_0 \mathcal{V}_N}, e^{-\Delta_1 \mathcal{V}_N}, \dots \right],$$

where the second equality is the Riemann-sum approximation of the time integral inside the bridge expectation: $\mathcal{V}_N$ is continuous, bounded below, and the bridge paths are continuous, so $\sum_i \Delta_i \mathcal{V}_N(B_{s_i}) \rightarrow \int_0^t \mathcal{V}_N(B_s) ds$ pointwise on paths and the integrands are dominated by e^{tc_V} times the free chain (bounded-below $\mathcal{V}_N$, by the displayed pointwise bound in Theorem 5.4), allowing dominated convergence under the (finite) free chain integral; the chain form of the bridge-times-endpoint integral is the semigroup (Chapman–Kolmogorov) property of $k^{(0)}$. Apply Lemma 5.3 at each level M : the marked-slice functional f_0 produces the factor $e^{i\varepsilon_N q \cdot (\boldsymbol{\eta}_0 + \mathbf{H}_p(0))}$ with $\mathbf{H}_p(0) = -p/2$, cancelling $e^{i\theta}$; the potential factors become $e^{-\Delta_i \mathcal{V}_N(\boldsymbol{\eta}_i + \mathbf{H}_p(s_i))}$. Undo the Riemann sum on the periodic side (same dominated convergence; $\mathbf{H}_p$ continuous), obtaining $e^{-S_N(\mathbf{H}_p)}$ times the periodic FK integral with the shifted potential, i.e. the numerator of (14). The denominator is the case $q = p = 0$. Finally (15) is the Appell (binomial) property of Wick powers: from $:u^4:_c = u^4 - 6cu^2 + 3c^2$,

$$\begin{aligned} :(\eta + H)^4:_c - :\eta^4:_c &= [4H\eta^3 + 6H^2\eta^2 + 4H^3\eta + H^4] - 6c[2H\eta + H^2] \\ &= 4H:\eta^3:_c + 6H^2:\eta^2:_c + 4H^3\eta + H^4, \end{aligned}$$

the last equality by $\eta^3 = :\eta^3:_c + 3c\eta$, $\eta^2 = :\eta^2:_c + c$, which cancels $-6c[2H\eta + H^2]$ identically (no residual Wick-constant term survives in the Wick-power form); the uniform bound on $\|\mathbf{H}_p\|_\infty$ follows from the *closed form* of the interpolant, not from (9) by the triangle inequality: the hyperbolic addition formula turns (9) into

$$H_k(s) = \frac{\rho_k}{2} \frac{\sinh(\gamma_N(k)(s - \frac{t}{2}))}{\sinh(\gamma_N(k)t/2)},$$

a one-line verification: expanding

$$\sinh(\gamma_N(s - \frac{t}{2})) = \sinh(\gamma_N s) \cosh(\frac{\gamma_N t}{2}) - \cosh(\gamma_N s) \sinh(\frac{\gamma_N t}{2})$$

and dividing by $\sinh(\gamma_N t/2)$ returns exactly $A_k \cosh(\gamma_N s) + B_k \sinh(\gamma_N s)$ with the A_k, B_k of (9). Hence $|H_k(s)| \leq |\rho_k|/2$ for every $s \in [0, t]$, because $|\sinh|$ is even and increasing and

$|s - \frac{t}{2}| \leq \frac{t}{2}$. (The same identity is recorded downstream as V2, Lemma 6.1; it is derived here from (9) alone, so no import is created.) The triangle inequality applied to (9) would give only $\frac{|\rho_k|}{2} (\cosh(\gamma_N s) + \coth(\frac{t\gamma_N}{2}) \sinh(\gamma_N s))$, which grows like $e^{\gamma_N t}$ and is useless in the regime $\gamma_N \sim 2/\varepsilon_N$; the cancellation in the closed form is what makes the bound uniform, with no t_0 -dependence. The inverse transform is an $\ell^2 \rightarrow \ell^\infty$ bounded map on $2r_N$ modes with the stated p -dependence. $\square$

Remark 5.5 (the periodic ensemble). $\mu_{N,t}^{\text{per}}$ is the finite product over $k \in \Gamma_N$ of the one-dimensional Gaussian processes on the circle of circumference t with covariance

$$c_\omega(s) = \frac{\cosh(\omega(\frac{t}{2} - |s|_{\mathbb{T}_t}))}{2\omega \sinh(\omega t/2)}, \quad \omega = \gamma_N(k), \quad (16)$$

which is the two-point function of the periodic Mehler chain: per mode, for $0 \leq s \leq t$ and $w = e^{-t\omega}$,

$$Z^{-1} \text{Tr}(e^{-(t-s)H} X e^{-sH} X) = \frac{e^{-s\omega} + w e^{s\omega}}{2\omega(1-w)} = c_\omega(s),$$

by the ladder computation $\langle n, X e^{-sH} X n \rangle$ -insertion and the geometric sums $\sum (n+1)w^n = (1-w)^{-2}$, $\sum n w^n = w(1-w)^{-2}$. Existence of a continuous version: $c_\omega(0) - c_\omega(s) \leq C_\omega |s|$, so $\mathbb{E}|\eta(s) - \eta(s')|^2 \leq C|s - s'|$ and, the process being Gaussian, $\mathbb{E}|\eta(s) - \eta(s')|^4 \leq 3C^2|s - s'|^2$; Kolmogorov–Chentsov applies. Its finite-dimensional marginals are the normalized periodic Mehler chains (both are Gaussian; the chain marginals are consistent by the semigroup property and have the two-point function above, computed operator-side), which is the form used in Lemma 5.3. To fix a normalization that the product description leaves implicit: the mode variables carrying (16) are the unitary DFT modes of the *rescaled* coordinates $\mathbf{X} = \varepsilon_N^{1/2} \boldsymbol{\varphi}$ of the import note (I1) in §1, not of $\boldsymbol{\varphi}$ itself, whose DFT modes carry c_ω/ε_N . Equivalently, in position space the $\boldsymbol{\varphi}$ -process has

$$\mathbb{E}[\varphi_x(s) \varphi_y(s')] = \frac{1}{2L} \sum_{k \in \Gamma_N} e^{ik(x-y)} c_{\gamma_N(k)}(s - s'),$$

which is the form matched verbatim by V2, Proposition 1.2(i); this was verified against the explicitly diagonalized chain at $r_N = 4, 8, 16$ to 4×10^{-14} .

Corollary 5.6 (free case closes the circle). *For $\lambda = 0$, (14) reads*

$$F_N^{(0)}(t; \xi) = e^{-S_N(\mathbf{H}_p)} \mathbb{E}[e^{i\varepsilon_N q \cdot \boldsymbol{\eta}(0)}] = e^{-S_N(\mathbf{H}_p)} e^{-\frac{\varepsilon_N^2}{2} q^\top C_N^{\text{per}}(0) q},$$

and by (16) ($s = 0$: $c_\omega(0) = \coth(\omega t/2)/(2\omega)$) and the mode Parseval identity this equals $\exp[-\frac{1}{4} Q_N^{(t)}(\xi)]$: the momentum half is (12) and the position half is

$$\frac{1}{4} (2r_N)^{-1} \sum_k \coth(t\gamma_k/2) \gamma_k^{-1} |\widehat{q}(k)|^2$$

— in exact agreement with Theorem 4.3. The two proofs are independent (operator ladder algebra vs. Gaussian chain calculus), so the agreement is a genuine internal cross-check of (14).

6 The continuum free functional (A3(a), continuum)

Let $H_{0,L} = d\Gamma(\mu)$, $\mu = (-\partial_x^2 + m^2)^{1/2}$, on the Fock space over $L^2(\mathbb{T}_L)$, with modes $k \in \Gamma_\infty = \frac{\pi}{L}\mathbb{Z}$ and $\gamma(k) = (m^2 + k^2)^{1/2}$. The model sheet's (2.13)–(2.16) fix $\phi_{M,x}$, R_M^∞ , the

scaling equation and β_M , but not the continuum Weyl data themselves; we therefore fix them here, once, and use them throughout Part II. The one-particle space is

$$h_{\text{ct}} := \{ \zeta = q + ip : \langle q, \gamma^{-1}q \rangle + \langle p, \gamma p \rangle < \infty \} = H^{-1/2}(\mathbb{T}_L) \oplus i H^{1/2}(\mathbb{T}_L),$$

not all of $L^2(\mathbb{T}_L; \mathbb{C})$: for $p \in L^2 \setminus H^{1/2}$ the operator $\Pi(p)$ is not densely defined and $W_{\text{ct}}(ip)$ is not unitary, so the larger space would not carry a Weyl system. On h_{ct} the symplectic form $\sigma_{\text{ct}}(\zeta, \zeta') = \text{Im} \int_{\mathbb{T}_L} \bar{\zeta} \zeta'$ is finite (it pairs $H^{-1/2}$ with $H^{1/2}$), and we set $W_{\text{ct}}(q + ip) := e^{i(\Phi(q) + \Pi(p))}$ with the CCR phase $\frac{1}{2}\langle q, p \rangle$, the continuum analogue of Proposition 2.1. Note the parametrisation is $\zeta = q + ip$, *without* the lattice's ε_M : with $\varepsilon_M q$ in place of q the Weyl relation would carry $\sigma_{\text{ct}}/\varepsilon_M$ and β_M would fail to be a $*$ -homomorphism. The free vacuum functional is

$$\omega_L^{(0)}(W_{\text{ct}}(\zeta)) = \exp \left[-\frac{1}{4} (\langle q, \gamma^{-1}q \rangle + \langle p, \gamma p \rangle) \right], \quad \zeta = q + ip.$$

With $q_\infty = \varepsilon_M^{1/2} \sum_x q(x) \phi_{M,x}$ one has $\int_{\mathbb{T}_L} q_\infty(y) e^{-iky} dy = \varepsilon_M^{1/2} P_\infty(\varepsilon_M k) \hat{q}_M(k)$ (using $\hat{\phi} = P_\infty$, free-rate (1.2)), so this functional evaluated on $R_M^\infty \xi$ is exactly the free-rate document's continuum vacuum (1.7)–(1.8); the two sides of the gate are thus normalized identically, and R_M^∞ is symplectic by orthonormality of $\{\phi_{M,x}\}_{x \in \Lambda_M}$. Finally R_M^∞ maps into h_{ct} : $R_M^\infty \xi$ has Fourier coefficients $\varepsilon_M^{1/2} P_\infty(\varepsilon_M k) \hat{q}_M(k)$ with $|P_\infty(\ell)| \leq C_D^{1/2} (1 + |\ell|)^{-1-\eta/2}$ (free-rate Lemma 2.1), so both $\langle q_\infty, \gamma^{-1}q_\infty \rangle$ and $\langle p_\infty, \gamma p_\infty \rangle$ converge — the latter needing $\sum_k \gamma(k)(1 + \varepsilon_M |k|)^{-2-\eta} < \infty$, which holds since $2 + \eta > 2$. Hence every Weyl generator the gate names is defined, and $\mathfrak{W}(\mathbb{T}_L)$ below is generated by $\{W_{\text{ct}}(\zeta) : \zeta \in h_{\text{ct}}\}$.

Theorem 6.1 (A3(a), continuum free). *$e^{-tH_{0,L}}$ is trace class for every $t > 0$, with $\text{Tr} e^{-tH_{0,L}} = \prod_{k \in \Gamma_\infty} (1 - e^{-t\gamma(k)})^{-1}$ (convergent), and for every finite-mode symbol — in particular for $R_M^\infty \xi$, whose mode amplitudes z_k are given by the continuum analogue of Lemma 4.1 with $\sum_k |z_k|^2 < \infty$ —*

$$\frac{\text{Tr}(e^{-tH_{0,L}} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr} e^{-tH_{0,L}}} = \exp \left[-\frac{1}{4} Q_\infty^{(t)}(R_M^\infty \xi) \right], \quad (17)$$

where $Q_\infty^{(t)}$ is the $\coth(t\gamma(k)/2)$ -weighted continuum covariance form (the vacuum form of the free-rate document, eq. (1.8), with the thermal weights inserted).

Proof. Here $H_{0,L} = \sum_k \gamma(k) a_k^* a_k$ (no zero-point subtlety: we use the number-operator normalization; ratios are unaffected). The occupation basis $\{|\mathbf{n}\rangle : \mathbf{n} \in \bigoplus_k \mathbb{N}_0 \text{ finitely supported}\}$ diagonalizes $e^{-tH_{0,L}}$ with eigenvalues $e^{-t \sum_k \gamma(k) n_k}$; summability: $\sum_{\mathbf{n}} e^{-t \sum \gamma n} = \prod_k (1 - e^{-t\gamma(k)})^{-1}$, finite because $\sum_k e^{-t\gamma(k)} < \infty$ ($\gamma(k) \geq c|k|$). For a Weyl operator with ℓ^2 mode amplitudes, $W = \bigotimes_k D_k(iz_k)$ in the incomplete-tensor-product sense over the vacuum stabilization, and $\langle \mathbf{n}, W \mathbf{n} \rangle = \prod_k \langle n_k, D_k(iz_k) n_k \rangle$: all but finitely many factors are $\langle 0, D(iz_k) 0 \rangle = e^{-|z_k|^2/2}$, and the product converges absolutely since $\sum |z_k|^2 < \infty$. Then

$$\text{Tr}(e^{-tH_{0,L}} W) = \sum_{\mathbf{n}} e^{-t \sum \gamma n} \prod_k \langle n_k, D_k(iz_k) n_k \rangle,$$

absolutely convergent with majorant $\text{Tr} e^{-tH_{0,L}}$ ($|\langle \mathbf{n}, D \mathbf{n} \rangle| \leq 1$, unitarity), so Fubini–Tonelli for countable products applies and the sum factorizes into

$$\prod_k \sum_{n_k} e^{-t\gamma n_k} \langle n_k, D_k n_k \rangle.$$

Lemma 4.2 evaluates each factor; taking the ratio with the partition function and inserting the continuum amplitude computation — identical to Lemma 4.1, with the normalization $(2r_N)^{-1} \sum_{\Gamma_N}$ replaced by $\varepsilon_M (2L)^{-1} \sum_{\Gamma_\infty}$ as in the free-rate document, Lemma 1.2 — gives (17). $\square$

7 The thermal fixed-coarse rate (A3(b))

Fix the coarse scale M , $\xi = \varepsilon_M q + ip$, and $t_0 > 0$. Write $w_t(\gamma) := \coth(t\gamma/2)$, $\bar{w} := \coth(t_0 m/2)$, and $x_0 := t_0 m/2$. Define the pulled-back thermal forms $Q_{r,M}^{(t)}$ and $Q_{\infty,M}^{(t)}$ exactly as in the free-rate document, Lemma 1.1/1.2, with the weight $w_t(\gamma_{M+r}(k))$ resp. $w_t(\gamma(k))$ inserted (V0 verified the two implementations of the pullback agree to 2×10^{-15} , i.e. at round-off level; v0_checkpoint.out, item [2], records 1.78×10^{-15} , and the value is not bit-reproducible across runs).

Lemma 7.1 (derivative bounds for the thermal weights). *Let $s_* := \sup_{x \geq x_0} x / \sinh^2(x) < \infty$. For all $\gamma \geq m$ and $t \in [t_0, \infty]$:*

$$\left| \frac{d}{d\gamma} [\gamma w_t(\gamma)] \right| = \left| w_t(\gamma) - \frac{t\gamma/2}{\sinh^2(t\gamma/2)} \right| \leq \bar{w} + s_*, \quad (18)$$

$$\left| \frac{d}{d\gamma} [\gamma^{-1} w_t(\gamma)] \right| = \left| -\frac{w_t(\gamma)}{\gamma^2} - \frac{t/2}{\gamma \sinh^2(t\gamma/2)} \right| \leq \frac{\bar{w} + s_*}{m^2}, \quad (19)$$

and $1 \leq w_t(\gamma) \leq \bar{w}$; at $t = \infty$, $w \equiv 1$ and both left-hand sides reduce to 1 and $\gamma^{-2} \leq m^{-2}$.

Proof. $\frac{d}{d\gamma} \coth(t\gamma/2) = -\frac{t}{2} \sinh^{-2}(t\gamma/2)$, so the two derivatives are as displayed. In (18), $w_t(\gamma) \leq \bar{w}$ (monotonicity of $\coth$ in both arguments, $t\gamma \geq t_0 m$) and $\frac{t\gamma}{2} \sinh^{-2}(\frac{t\gamma}{2}) = x / \sinh^2 x|_{x=t\gamma/2} \leq s_*$ since $x \geq x_0$. In (19), the first term is $\leq \bar{w}/m^2$ and the second is $\frac{t/2}{\gamma \sinh^2} = \frac{x}{\gamma^2 \sinh^2 x} \leq s_*/m^2$. $\square$

Theorem 7.2 (A3(b): thermal rate, uniform in t). *Let $\eta = 2(1 - \log_2 \sqrt{3})$, $\theta = 2\eta/(5 + \eta)$, and let r_* , J_r , C_D , A_q , A_p , b , κ , c_0 be as in the free-rate document, §4. Then for all $r \geq r_*$ and all $t \in [t_0, \infty]$,*

$$|Q_{r,M}^{(t)}(\xi) - Q_{\infty,M}^{(t)}(\xi)| \leq C_Q^{\text{th}} 2^{-\theta r}, \quad \left| F_{M+r}^{(0)\text{-side}} - F_{\infty}^{(0)\text{-side}} \right| \leq \frac{1}{4} C_Q^{\text{th}} 2^{-\theta r}, \quad (20)$$

where the second display compares the two thermal Weyl functionals of Theorems 4.3 and 6.1 on the embedded generator, and

$$\begin{aligned} C_Q^{\text{th}} &:= C_{\text{low}}^{\text{th}} (1 + b^{-1})^5 + \frac{2^\eta C_*^{\text{th}}}{\pi \eta}, \\ C_{\text{low}}^{\text{th}} &:= \frac{3\varepsilon_M^3}{2L} \left\{ A_q^2 \left(\frac{(\bar{w} + s_*)\kappa^4}{24m^3} + \frac{c_0 \bar{w} \kappa^2}{m} \right) + A_p^2 \left(\frac{(\bar{w} + s_*)\kappa^4}{24m} + c_0 \bar{w} (m + \kappa) \kappa^2 \right) \right\}, \\ C_*^{\text{th}} &:= \bar{w} C_* = 2 \bar{w} C_D \left(\frac{A_q^2}{m} + A_p^2 (m + \varepsilon_M^{-1}) \right). \end{aligned} \quad (21)$$

Proof. Follow the proof of the free-rate Theorem 4.1 step by step; only the weights change. *Low momenta* ($|n| \leq J_r$): the two integrand errors are now, by the triangle splits

$$|w(\gamma_N)\gamma_N|P_r|^2 - w(\gamma)\gamma|P_\infty|^2| \leq |w(\gamma_N)\gamma_N - w(\gamma)\gamma| |P_r|^2 + w(\gamma)\gamma ||P_r|^2 - |P_\infty|^2|,$$

and its q -analogue; the mean value theorem with Lemma 7.1 and the imported dispersion bound $|\gamma_N - \gamma| \leq \varepsilon_N^2 k^4 / (24m)$ (free-rate Lemma 3.1) give

$$|w(\gamma_N)\gamma_N - w(\gamma)\gamma| \leq (\bar{w} + s_*) \frac{\varepsilon_N^2 k^4}{24m}, \quad |w(\gamma_N)\gamma_N^{-1} - w(\gamma)\gamma^{-1}| \leq \frac{\bar{w} + s_*}{m^2} \frac{\varepsilon_N^2 k^4}{24m},$$

while the imported filter-tail bound (free-rate Lemma 2.2) is multiplied by $w(\gamma)\gamma \leq \bar{w}(m + \kappa(1 + J))$ resp. $w(\gamma)/\gamma \leq \bar{w}/m$. Summing over $|n| \leq J_r$ exactly as in the source (the counting

bounds (4.16) are unchanged) yields $E_{\text{low}} \leq 4^{-r} C_{\text{low}}^{\text{th}} (1 + J_r)^5$. *High momenta*: the envelope estimate (free-rate Lemma 2.1) is unchanged and the summand acquires the factor $w \leq \bar{w}$, so C_* becomes $C_*^{\text{th}} = \bar{w} C_*$ and the integral-comparison tail (4.20)–(4.21) goes through verbatim. Balancing the exponents as in the source ($2 - 5\alpha = \alpha\eta = \theta$) proves the first display of (20); the second follows from $|e^{-x/4} - e^{-y/4}| \leq \frac{1}{4}|x - y|$ ($x, y \geq 0$). All bounds are uniform on $t \in [t_0, \infty]$, with the $t = \infty$ case reducing to the printed vacuum theorem (weights $\equiv 1$, s_* -terms replaced by the original ones, and $C^{\text{th}} \geq C^{\text{vac}}$). $\square$

Remark 7.3 (V0 acceptance). The measured slopes at $t \in \{1, 2, 4\}$ and $t = \infty$ coincide to four decimals (drift 0.0000), consistent with the t -uniform bound (20); the measured decay $2^{-1.011r}$ for the smooth V0 symbol is faster than the worst-case $2^{-\theta r}$, as expected.

8 A2(b): the continuum statement and its input clause

Definition 8.1 (A2(b), final form). Let K_L be the continuum torus Hamiltonian in the A4b convention. The continuum marked-slice representation is the statement: for $t > 0$ and every coarse symbol ξ ,

$$\frac{\text{Tr}(e^{-tK_L} W_{\text{ct}}(R_M^\infty \xi))}{\text{Tr } e^{-tK_L}} = e^{-S_\infty(H_p)} \frac{\mathbb{E}_{\mu_{t,L}^{\text{per}}}[e^{i\Phi(J_q)} e^{-\mathcal{U}(\eta + H_p)}]}{\mathbb{E}_{\mu_{t,L}^{\text{per}}}[e^{-\mathcal{U}(\eta)}]},$$

with S_∞ , H_p the mode-wise objects of Lemmas 5.2–5.3 for the dispersion $\gamma(k)$ (the free ingredients are proved: Theorem 6.1 and the per-mode statements apply verbatim), J_q the sharp-time position source (admissible: Lemma 2.1 of the scoping note concerns momentum only), and $\mathcal{U}$ the continuum interaction.

Remark 8.2 (input clause and amendment). The only unproved ingredient in Definition 8.1 is the continuum Feynman–Kac/trace formula for e^{-tK_L} on the space–time torus (the $\lambda = 0$ case is Theorem 6.1; the Gaussian shift lemma is mode-wise and dimension-free). Accordingly, Statement A1 of the scoping note is *extended by the clause* (A1-iv): “ e^{-tK_L} admits the periodic Feynman–Kac representation over $\mathbb{T}_t \times \mathbb{T}_L$, compatible with cylinder approximation.” A2(b) is then a theorem given A1(i)–(iv), by the identical chain of §5: kernel identity (the spectral/Fubini argument of Theorem 3.3 uses only trace-class, A1(ii)), shift identity (mode-wise, proved), and Riemann/dominated convergence (uses $e^{-\mathcal{U}} \in L^1$, part of A1(iv)’s package). The amendment is recorded in `strategy.md`; A1 including (A1-iv) is the V3 deliverable, with Høegh-Krohn’s finite-temperature construction as the import candidate for precisely this clause.

9 Acceptance and claim ledger

claim	status	ground
Schrödinger action of W_N , CCR phase $\frac{1}{2}\varepsilon_N q \cdot p$	proved	Prop. 2.1; V0 corner checks, residuals $\leq 4 \times 10^{-15}$
Trace class of $e^{-tK_{N,L}}$, $Z_N(t) \leq e^{tcv} Z_N^{(0)}(t)$	proved	Prop. 3.1
Kernel identity for the marked-slice trace	proved	Thm. 3.3 (eigenfunction bounds, Fubini)

claim	status	ground
Free thermal functional, lattice (coth form)	proved	Thm. 4.3; V0: ED residual 6×10^{-10} ; position-space cross-check passed at tolerance 10^{-9}
Single-mode thermal displacement	proved	Lem. 4.2 (elementary Laguerre summation)
Harmonic jump interpolant; exact three-term recursion; action $= \frac{\omega}{4} \coth(\frac{\omega t}{2}) \rho^2$	proved	Lem. 5.2
Exact Gaussian shift identity at every partition	proved	Lem. 5.3 (Mehler-chain harmonicity is exact)
Interacting mean-shift representation (lattice); phase cancellation; Wick-binomial shift	proved	Thm. 5.4
Free case agreement of the two routes	proved	Cor. 5.6
Continuum free thermal functional	proved	Thm. 6.1 (countable-product Fubini)
Thermal fixed-coarse rate, uniform on $[t_0, \infty]$	proved	Thm. 7.2; imports (I2); V0 slope drift 0.0000
A2(b) continuum interacting representation	conditional	Def. 8.1 + clause (A1-iv); deferred to V3 (Rem. 8.2)

Phase V1 gate: A2(a) and A3 are closed; A2(b) is in final conditional form with its single input clause isolated and reassigned to V3. The V0 acceptance criteria (formulas match the tables to rounding) are met by Theorems 4.3, 6.1, and 7.2.